

Mathematics (OG)

Basic Mathematics

Lecture-06

Concepts Covered in the Class:

- Domain Conditions of Exponential and Logarithmic Functions
- Graphical Behavior of Logarithmic and Exponential Functions
- Inverse Symmetry and Reflections across $y = x$
- Exponential and Logarithmic Inequalities
- Sum and Product of Roots in Quadratic Equations of Non-Linear Forms

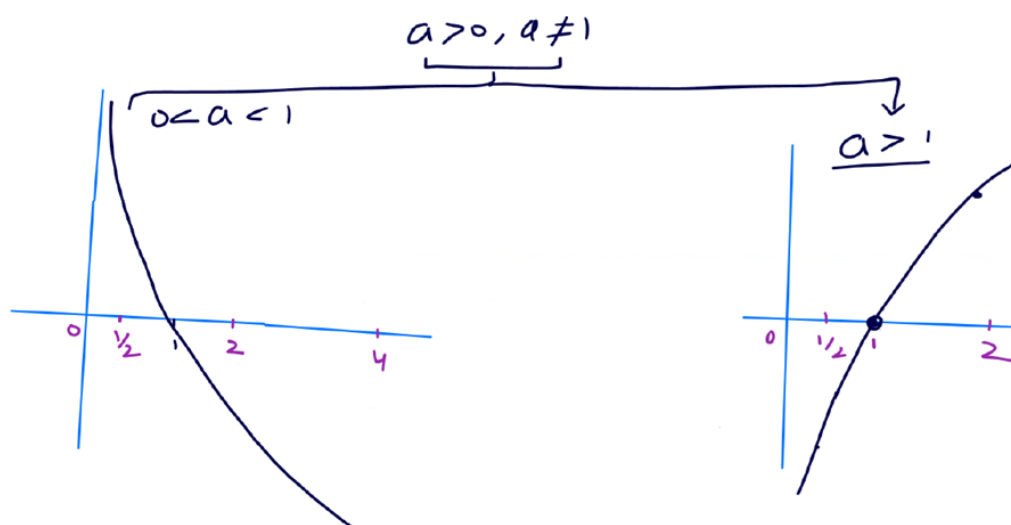
Topics To Be Covered

- The exponential function is defined as $y = a^x$, where the base must satisfy $a > 0$ and $a \neq 1$, and the exponent $x \in \mathbb{R}$, yielding a positive output $y > 0$.
- The logarithmic function $y = \log_a x$ is the inverse of the exponential function, requiring $x > 0$, $a > 0$, and $a \neq 1$, with the output $y \in \mathbb{R}$.



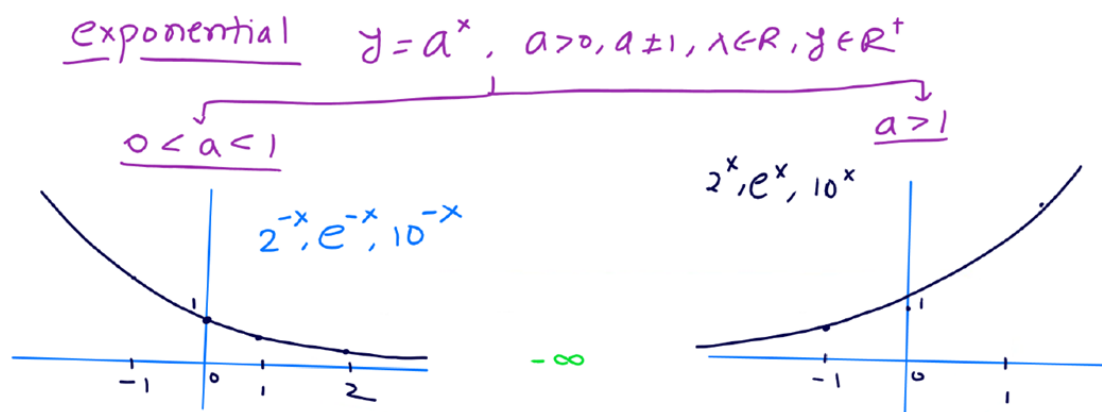
log

- The graph of the logarithmic function $y = \log_a x$ is strictly decreasing when the base is in the range $0 < a < 1$.
- The graph is strictly increasing when the base $a > 1$, and all curves of this form pass through the point $(1, 0)$ on the x-axis.



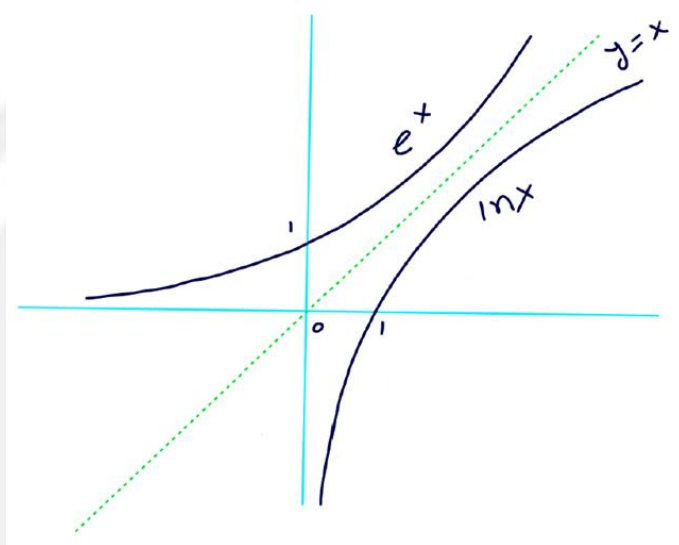
exponential

- The exponential function $y = a^x$ is strictly decreasing for base values $0 < a < 1$ and strictly increasing for base values $a > 1$.
- These curves always stay entirely above the x-axis ($y > 0$) and cross the y-axis at the point $(0, 1)$.



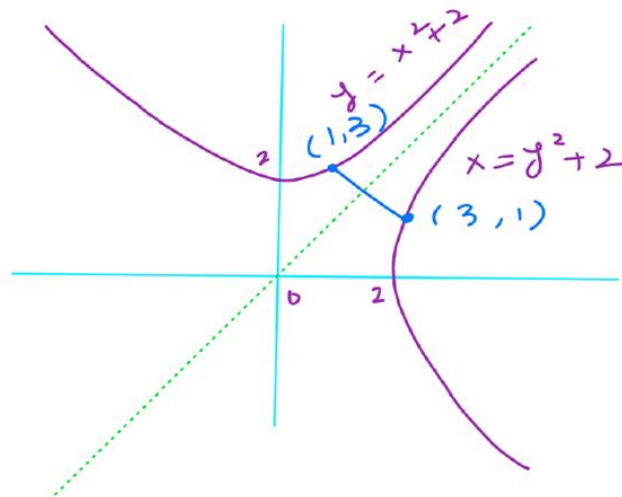
$$y = \ln x$$

- The natural exponential function $y = e^x$ and the natural logarithmic function $y = \ln x$ are inverse operations of each other.
- Geometrically, inverse functions are symmetric reflections across the diagonal identity line $y = x$.



$$y = x^2 + 2$$

- Swapping x and y in an equation, such as changing $y = x^2 + 2$ to $x = y^2 + 2$, produces curves that are symmetric reflections across the line $y = x$.
- Any coordinate point (a, b) lying on the original curve corresponds to the point (b, a) on its reflected inverse curve.



EXPONENTIAL INEQUALITIES

- For inequalities involving exponential or logarithmic terms, the inequality sign remains the same when the base is greater than 1 ($a > 1$).
- The inequality sign reverses when the base lies between 0 and 1 ($0 < a < 1$).

$$A(\log_a x)^2 + B(\log_a x) + C = 0$$

- To find the product of roots ($x_1 \cdot x_2$) for a quadratic-style logarithmic equation, use the sum of roots property: $\log_a(x_1 \cdot x_2) = -B/A \Rightarrow x_1 \cdot x_2 = a^{-B/A}$.
- To find the sum of roots ($x_1 + x_2$) for a quadratic-style exponential equation $A(a^x)^2 + B(a^x) + C = 0$, use the product of roots property: $a^{x_1+x_2} = C/A \Rightarrow x_1 + x_2 = \log_a(C/A)$.